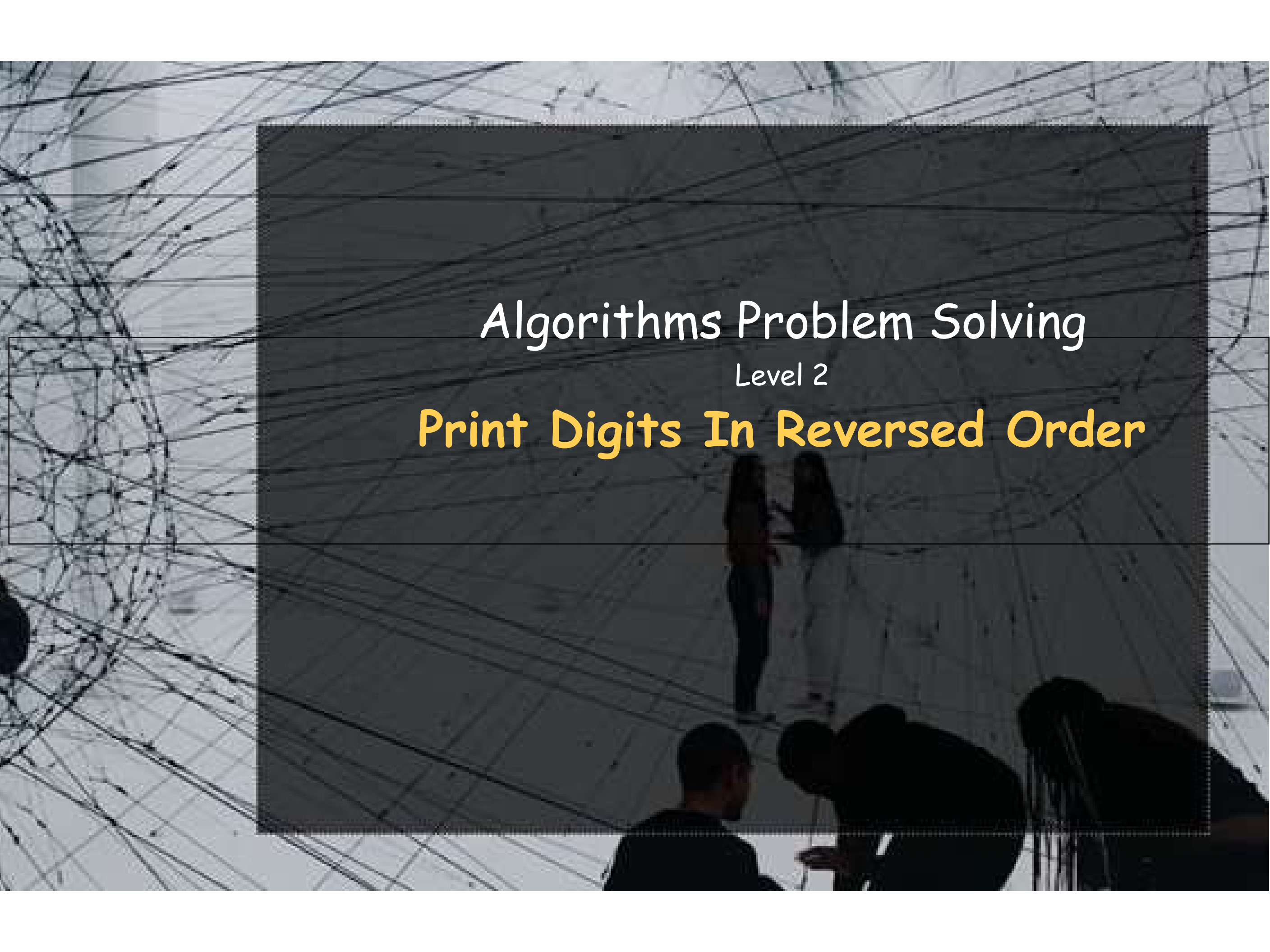


Algorithms Problem Solving

Level 2

Print Digits In Reversed Order



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Definition

◆ Reversing the digits of a number means reading and printing its digits starting from the last digit to the first digit.

Explanation

- ◆ ``ReadNumber()`` reads a positive number from the user using a ``do-while`` loop.
- ◆ ``PrintResult()`` uses a ``while`` loop to extract the last digit of the number one digit at a time.
- ◆ ``N % 10`` gives the last digit of ``N``.
- ◆ ``N = N / 10`` removes the last digit from ``N``.
- ◆ The extracted digit is printed before moving to the next digit.

Steps in main()

1. Read a positive number N .
2. Get the last digit using $N \% 10$.
3. Print the extracted digit.
4. Remove the last digit using $N / 10$.
5. Repeat while $N > 0$.

💡 Example

◆ Input:

`1234`

◆ First: ` $1234 \% 10 = 4$ `

◆ Then: ` $1234 / 10 = 123$ `

◆ Next: ` $123 \% 10 = 3$ `

◆ Then: ` $123 / 10 = 12$ `

◆ Next: ` $12 \% 10 = 2$ `

◆ Then: ` $12 / 10 = 1$ `

◆ Finally: ` $1 \% 10 = 1$ `

➡ Result

`4`

`3`

`2`

`1`

Note

- ◆ ``% 10`` is used to extract the last digit of a number.
- ◆ Integer division by ``10`` removes the last digit.
- ◆ The variable ``Sum`` is declared but never used in this program.

⊘ Warning

- ◆ The program prints each digit on a separate line rather than on the same line.
- ◆ Since the input must be positive, `0` cannot be entered.



Comparison

▨ `N % 10`

- ✓ Extracts the last digit.
- ✓ Example: `1234 % 10 = 4`.

▤ `N / 10`

- ✓ Removes the last digit when using integer division.
- ✓ Example: `1234 / 10 = 123`.

✦ Summary

- ✓ Use ``% 10`` to get the last digit.
- ✓ Use ``/ 10`` to remove the last digit.
- ✓ Repeat until the number becomes ``0``.
- ✓ The digits are printed in reversed order.

Thank you 😊